



Impact Study Methodology:

Quasi-experimental treatment-on-the-treated impact research

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Summary

Zearn regularly partners with school districts and states to conduct quasi-experimental impact studies aimed at estimating the causal effects of Zearn Math on student achievement among students who consistently complete lessons on the platform, defined as completing lessons each week throughout the school year. This paper outlines the methodology used for such studies, which is one of several ways Zearn measures its impact and effectiveness.

To approximate an experimental design, Zearn researchers apply coarsened exact matching (CEM) with an optimal matching algorithm (described in the “Quasi-experimental matching techniques” section below). This method creates a comparison group of students with no Zearn Math usage that is baseline equivalent to consistent Zearn Math users on prior-year or beginning-of-year math assessment scores, and similar on prior-year or beginning-of-year ELA assessment scores, as well as other academic and demographic covariates. By reducing the influence of confounding variables, CEM allows researchers to more accurately isolate the treatment effect and draw causal inferences.

The analyses use a treatment-on-the-treated (TOT) framework to assess the impact of Zearn Math on student achievement among students who consistently complete lessons each week throughout the school year. A dose-response model is employed to examine impact at the following thresholds: 1-1.9, 2-2.9, and 3+¹ unique on- or above-grade lessons per week. Differences between matched treatment and control groups are assessed using linear regression models, which control for prior-year or beginning-of-year math and ELA scores, as well as key demographic and academic characteristics, to reduce bias in the estimated treatment effects.

The TOT analytic framework is a statistical method commonly used in observational studies to estimate the causal effect of a treatment on the subgroup of individuals who actually receive it. The TOT estimate focuses specifically on those who are exposed to the intervention (Rubin, 1974). This makes the TOT approach particularly valuable in real-world contexts, such as Zearn Math implementations, where all students in a school or district are rostered and given access to the platform, regardless of their actual intent to use it. By concentrating on the students who actively engage with the platform, the TOT framework provides a more meaningful measure of the intervention's impact.

¹ Three or more on- or above-grade lessons per week represents Zearn’s recommended usage for maximum impact.

Zearn Math Background

Zearn was initially launched in 2016 as a K-5 core curriculum. Over time, its primary use evolved into a supplemental digital math program for math instruction featuring digital lessons and workbooks. The program expanded to K-8 in 2022 with the progressive release of middle school digital lessons (grades 6-8) beginning in 2019.

Each digital lesson takes approximately 30 minutes and includes fluency games, videos led by on-screen teachers with pause points to solve math problems, and a closing mastery-based quiz that students must achieve 100% on in order to open up the next lesson in the sequence. Each grade, K-8, includes 120 digital lessons.

Research Questions

Zearn's impact studies address the following questions:

1. What is the effect of Zearn Math on student achievement on interim and/or state assessments for students who use Zearn consistently² compared to those who do not?
2. What is the effect of Zearn Math on student achievement on interim and/or state assessments for subgroups³ of students who use Zearn with fidelity (3+ lessons-per-week) compared to those who do not?

Research Methodology

Treatment group

Impact studies measure the treatment-on-the-treated (TOT) effects of Zearn Math usage on student achievement, by comparing a similar group of consistent Zearn Math users who complete lessons each week of the school year to non-Zearn users on standardized math assessment scores.

² Consistent use of Zearn Math is defined as completing at least one lesson-per-week. Impact studies evaluate the effectiveness of Zearn Math for students based on three usage levels: 1-1.9 lessons-per-week, 2-2.9 lessons-per-week, and 3 or more lessons per week.

³ Student subgroups include prior-year achievement levels, grade-level, economic disadvantage, gender, race/ethnicity, students with disabilities, gifted, multilingual learners, and chronically absent.

A dose-response model is used to examine academic growth in standardized assessment scores for three distinct treatment groups to capture the incremental effects of Zearn usage on student achievement:

- Students who complete 1-1.9 on- or above-grade-lessons-per-week, the annual equivalent of 30-59;
- Students who complete 2-2.9 on- or above-grade-lessons-per-week, the annual equivalent of 60-89; and
- Students who complete 3+ on- or above-grade-lessons-per-week, the annual equivalent of 90+ lessons. This is the recommended Zearn usage threshold for maximum impact.

Treatment group sample sizes for Zearn’s impact analyses exceed the 350 student minimum required by ESSA to qualify as an evidenced-based intervention.

Control group

The control group is composed of matched students from the same population with no Zearn Math usage, identified as completing <1 lesson-per-week, the annual equivalent of <30 lessons-per-year who are similar to the treatment group students on prior-year or beginning-of-year math and ELA assessment scores, and covariates including district, school, grade, race/ethnicity, gender, economic disadvantage, multilingual learner status, chronic absenteeism, special education status, and gifted status.⁴

Control groups meet the What Works Clearinghouse standard for baseline equivalence to treatment students, with or without adjustment, on prior-year or beginning-of-year math assessment scores⁵ (What Works Clearinghouse, 2022)⁶.

Quasi-experimental matching techniques

The quasi-experimental method of Coarsened Exact Matching (CEM) is used to select a control group of students with no Zearn Math usage that are in the same grade, baseline equivalent to consistent Zearn Math users on prior-year or beginning-of-year math assessment scores, and similar on prior-year or beginning-of-year ELA assessment scores and relevant demographic and academic factors. Because an individual cannot simultaneously receive and not receive the intervention during the same year,

⁴ The specific academic and demographic characteristics selected for match are based on data availability from district or state partners.

⁵ Academic scores are state assessment or validated third-party interim scores from the spring prior to the study year or the fall of the study year in districts that implement standardized fall assessments.

⁶ According to WWC, baseline differences < .05 of a standard deviation satisfy baseline equivalence without adjustment. Differences < .25 of a standard deviation satisfy baseline equivalent with statistical adjustment. Difference-in-difference is an acceptable statistical adjustment (2022).

CEM approximates this scenario by matching treated students with non-treated students who are as similar as possible across these covariates (Ling et al., 2020).

Zearn’s impact analyses use a two-step CEM method, with optimal matching, to create a control group of similar students. CEM is a technique that simulates block sampling by matching students on covariates rather than propensity scores to create balanced, baseline-equivalent samples (Blackwell et al., 2009; Imai et al., 2009; Iacus et al., 2011; King et al., 2011; King & Nielsen, 2019). The goal of matching is to identify a control group that closely mirrors the treatment group, differing primarily in their usage of Zearn Math. This approach allows the analysis to focus on the TOT effect, isolating the impact of consistent Zearn Math usage.

By matching on potentially confounding covariates⁷, CEM creates comparable groups of students who did and did not receive the intervention, allowing for the isolation of the treatment effect, independent of confounders (Stuart, 2008; Iacus et al., 2011; King & Nielsen, 2019). As with any matching method, the success of CEM depends on the inclusion of a robust set of observable covariates that adequately model the selection process and reduce bias in the estimates.

Consistent Zearn Math users are matched with up to ten non-users on starting math and English Language Arts (ELA) achievement scores⁸ along with demographic variables using a two-step process. Using CEM, treatment students are put into matching strata with control students that are in the same grade and have proximate prior-year or beginning-of-year scores. Scores may be in the form of scale scores or national percentiles and are selected based on the metric used by the district for internal goals and reports. Generally score distance is set at a maximum of 5-10 points on both math and ELA. Score distance is selected to achieve closeness without losing a significant portion of the sample, thereby potentially making findings more biased and less generalizable. Closeness on math is prioritized over ELA when necessary because of its greater relevance to outcomes.

Within strata, treatment students are matched to control students with whom they share a majority of characteristics on factors such as district, school, gender, race/ethnicity, students with disabilities,

⁷ For example, in education research, an individual’s school or district likely impacts whether or not they receive a curriculum or classroom intervention. Prior performance might also be relevant if intervention is aimed specifically at low-performing or high-performing students. Language ability might be relevant if it is an intervention primarily provided to fluent English speakers. Demographic and socioeconomic factors are also frequently associated with whether or not an individual participates in a particular program or intervention.

In addition to shaping whether or not an individual receives an intervention, these covariate values are associated with differences in academic performance. Therefore in order to isolate an intervention as the “cause” of differences in academic achievement, quasi-experimental methods balance these factors so their impact is similar between treatment and control.

⁸ Achievement scores are state or externally validated third-party interim assessment scores from the spring prior to the study year or the fall of the study year in districts that implement standardized fall assessments.

multilingual learner status, gifted status, chronic absenteeism, and economic disadvantage.

Matches are identified using a custom Python application built in Hex that allows specification of maximum score distances between treatment and control students for math and ELA prior-year or beginning-of-year scores, as well as demographic “hard” matching criteria that matches must share, and “soft” matching criteria for which students must share a specified minimum number of characteristics.

Consistent Zearn Math users are matched with up to ten non-users, without replacement, using an algorithm that preferences, in descending order: leaving as few users as possible unmatched; minimizing the maximum number of controls matched to a single treatment user; and closeness of match. The algorithm used is an extension of Hansen and Klopfer’s network-flow based algorithm and the solution is a global optimum (2006).

Optimal matching through this Hex application offers greater precision than traditional CEM, where strata are fixed, limiting matches to treatment and control students within predefined strata, even when closer matches may exist in adjacent strata. In contrast, the Hex application allows treatment students to be matched to any student within the sample space that meets the specified criteria, offering more flexibility and improved matching accuracy.

Zearn’s matching method utilizes Bertsekas’ auction algorithm to produce combinatorial optimization such that treatment individuals are matched to others closest to them in the control pool. When controls are the best-fit match for more than one treatment individual, the pairing goes to the individual from whom the next best pairing is the farthest (1981; Rosenbaum, 2020).⁹

To be included in the pool of treatment or control matches, students must have prior-year or beginning-of-year ELA and math assessment scores, end-of-year math assessment scores, all hard match criteria, such as grade-level, and non-missing data for at least the minimum number of required academic and demographic characteristics for match. Cases with missingness in any of the mandatory match criteria are not included in the sample.

If a treatment student has no match within their grade and score strata with whom they share a majority of characteristics, they are excluded from the treatment group. The caliper that limits match difference is selected to maximize inclusion in the sample and prevent biasing the sample through

⁹ In other words, if Control Student A is the best match for Treatment Student 1 and Treatment Student 2, sharing 6 out of 7 characteristics with each, Control Student A can still only be matched with either Treatment Student 1 or Treatment Student 2. If the next best match for Treatment Student 1, Control Student B, shares 4 characteristics, and the next best match for Treatment Student 2, Control Student C, shares 5 characteristics, then Treatment Student 1 would be matched with Control Student A and Treatment Student 2 would be matched with Control Student C. In this way, the algorithm of optimal matching balances the closeness of any individual match with its impact on the closeness of the overall group match.

uneven patterns of exclusion, while ensuring similarity between groups.

Once CEM has identified a matched control group for consistent Zearn Math users, baseline equivalence is measured using the Hedge's g formula (described below in Models 2-4) on prior-year or beginning-of-year math assessment scores.

Dose-response analysis

In order to identify Zearn Math's impact across a range of "typical use" for students who consistently engage with the platform, impact studies structure effects as a dose-response, identifying effects among students who use at the recommended threshold of 3+ lessons per week, as well as lower thresholds of 1-1.9 lessons-per-week, and 2-2.9 lessons-per-week.

Matching between treatment and control students is conducted separately for each treatment threshold using Coarsened Exact Matching (CEM), with consistent demographic and score calipers applied across thresholds. Each treatment student is allowed up to ten control matches selected with the matching preference algorithm described above. The results from the dose-response analysis reflect the average achievement differences between students at each treatment threshold and their matched counterparts from the control group.

Since matching is performed independently for each threshold, the control groups for the three dosage levels are not identical. Instead, each control group represents the best possible match for students at that specific threshold. This approach preserves the quasi-experimental integrity of the analysis by ensuring that comparisons are made between similar treatment and control students, even if demographic attributes or baseline proficiencies vary across dosage groups.

Matching is performed without replacement within each threshold, but all thresholds draw from the same pool of control students. As a result, a student may appear as a control for multiple treatment groups if they are the best match for students at more than one threshold. Ultimately, the outcomes reflect findings from three separate quasi-experiments, with standardized matching criteria applied consistently across all three analyses.

Models are run separately for each dosage threshold with control students weighted as the inverse-proportion of their matching ratio to treatment students. Results illustrate the difference in end-of-year assessment scores between each dosage threshold sample and its baseline equivalent control sample. Differences across these analyses are compared in terms of treatment and control differences in scale score or percentile rank and in the relative effect sizes measured in each analysis.

Linear regression models for continuous outcomes

When comparing consistent Zearn Math users with a similar group of non-users, the appropriate linear regression models are selected based on whether the assessment used is vertically scaled. Vertically scaled assessments link scores across grade levels on a common scale, allowing for direct analysis of multi-grade samples. In contrast, non-vertically scaled assessments have distinct scales for each grade level¹⁰, requiring additional adjustments.

For multi-grade samples using non-vertically scaled assessments, regression models must include grade-level dummy variables to account for scaling differences. Alternatively, scores can be converted to z-scores standardized around state (or sample) means and standard deviations for each grade level. Given that many state summative assessments are not vertically aligned, using either grade-level controls on unstandardized data or standardizing scores ensures more accurate interpretation in dose-response and subgroup analyses.

When test scores are standardized, model coefficients represent the effect size of the treatment, rather than the raw scale scores. Results are typically reported as effect sizes, converted into Hedge's *g*. To provide further context, z-scores can be translated into changes in percentile rankings within the sample, illustrating the equivalent movement in percentiles between consistent Zearn Math users and non-users.

Linear regression models are used to analyze the impacts of Zearn Math on student achievement. Analysis is conducted in STATA using the *regress* command.

Student-level academic gains are modeled as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon \quad (\text{Model 1})$$

In this model, *Y* represents end-of-year assessment scores, while $\beta_1 \dots \beta_n$ represent student-level characteristics, including treatment or control group membership, prior-year or beginning-of-year math and English Language Arts (ELA) assessment scores, gender, race/ethnicity, special education status, multilingual learner status, gifted status, economic disadvantage, chronic absenteeism, and any other relevant district- or state-specific demographic and academic variables¹¹. Grade level

¹⁰ For instance, on a non-vertically scaled assessment, an increase of 7 scale score points in grade 3 may not be equivalent to the same increase in grade 5 due to differences in content, rigor, and expected growth across grade levels. Without controlling for grade-level variations or standardizing the scores, analyzing multi-grade samples may obscure the true treatment effect and complicate overall interpretation.

¹¹ When assessment data has been standardized to accommodate non-vertical scaling, *Y* and all assessment covariates are standardized.

covariates are included in the model only when assessment data is neither vertically scaled nor standardized.

In addition to measuring overall differences in student achievement between consistent Zearn Math users and matched non-users, impact analyses also examine achievement differences among specific student subgroups between fidelity users and non-users. These subgroups include students at each prior-year or beginning-of-year achievement level, as well as demographic and academic characteristics such as gender, race/ethnicity, students with disabilities, multilingual learner status, gifted status, economic disadvantage, grade-level, chronic absenteeism, and other district- or state-specific groups or academic programs. Subgroup analyses may be limited by sample size or data availability.

Because treatment and control matches are allowed to differ by a small number of demographic and academic characteristics, subgroups may not meet baseline equivalence on prior-year or beginning-of-year math assessment scores even if the overall matched sample does. Therefore, baseline equivalence is measured separately for each subgroup using the Hedge's g formula (described below in Models 2-4). Only subgroups that meet baseline equivalence and have at least 150 treatment students¹² are analyzed for Zearn Math's impact.

To analyze Zearn Math's impact within baseline equivalent subgroups, Model 1 (described above) is applied separately to the subsample of students in each subgroup.

Effect sizes for all analyses are calculated using the Hedge's g formula, which divides the difference in means between the treatment and control groups by the pooled, weighted standard deviations, and then multiplies by a small-sample correction factor (w). While similar to Cohen's d , Hedge's g incorporates this correction factor to produce unbiased effect size estimates, particularly in small samples (Hedges, 1981).

The general formula for Hedge's g is modeled as:

$$g = \frac{\omega\beta}{SD_p} \quad (\text{Model 2})$$

Where pooled standard deviation SD_p are calculated as:

$$SD_p = \sqrt{\frac{(n_t - 1)SD_t^2 + (n_c - 1)SD_c^2}{n_t + n_c - 2}} \quad (\text{Model 3})$$

¹²Implementation studies focusing on a limited number of districts, rather than statewide initiatives, may report results for groups of 100 or more students.

Such that n_t is the treatment sample size and n_c is the control sample size. The small-sample correction factor (w) is given by:

$$\omega = 1 - \frac{3}{4df-1}$$

And degrees of freedom (df) is calculated as:

$$df = n_t + n_c - 2 \quad (\text{Model 4})$$

Effect size interpretations are based on Matt Kraft's 2019 updated benchmarks, developed from the distribution of 1,942 effect sizes from 747 RCTs evaluating education interventions. Kraft recommends that <0.05 is Small, 0.05 to less than 0.20 is Medium, and 0.20 or greater is Large and notes the median impact across education RCTs reviewed was .10.

Because treatment students are matched to multiple control students who fall within matching calipers, control students are weighted as the inverse of the matching proportion using the *aweight* command in STATA. This rebalances estimates to resemble 1:1 matched pairs.

Logistic regression models for dichotomous outcomes

In some instances, assessment results include dichotomous outcomes in which state and district partners are interested in, such as students meeting growth targets on interim benchmarks. In these cases, analysis of the probability of students achieving outcomes with or without consistent Zearn usage is measured using a logistic regression model. Analysis is conducted in STATA using the *logistic* command.

Probabilities of students achieving outcomes are modeled as follows:

$$p(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (\text{Model 5})$$

In this model, $p(y=1 | x)$ is the probability that a student achieves a dichotomous assessment outcome, e is the base of the natural logarithm, and $\beta_1 \dots \beta_n$ represent student-level characteristics, including treatment or control group membership, prior-year or beginning-of-year math and English Language Arts (ELA) assessment scores, grade-level, gender, race/ethnicity, special education status, multilingual learner status, gifted status, economic disadvantage, chronic absenteeism, and any other relevant

district- or state-specific demographic and academic variables. Grade level coefficients are included in the model when assessment data is not vertically scaled.

Per What Works Clearinghouse guidance for analysis of dichotomous outcomes, effect sizes are reported as the Cox Index (2022):

$$d_{cox} = \omega(\ln(\frac{p_t}{1-p_t}) - \ln(\frac{p_c}{1-p_c}))/1.65 \quad (\text{Model 6})$$

In this model, p_t is the prevalence rate of an outcome for a student observed in the treatment group; p_c is the prevalence rate of an outcome for a student observed in the comparison group; and omega (ω) is the small sample size correction:

$$\omega = 1 - \frac{3}{4N-9} \quad (\text{Model 7})$$

Where N is the total sample size ($n_t + n_c$) such that n_t is the treatment sample size and n_c is the control sample size.

Prior analytic models

Impact analyses conducted prior to 2024 utilized CEM to identify a group of control students baseline equivalent to students who consistently used Zearn Math, with the same set of matching criteria and calipers detailed above. However, there were 3 major methodological differences:

1. **Matched pairs:** In studies conducted prior to 2024, each treatment student was matched to only one control student. Because the studies compared matched pairs, no weighting was applied to balance sample proportions. Revised methods allow each treatment student to match up to ten control students, increasing the precision of estimates by making it less likely that outliers or atypical control units disproportionately affect the results.
2. **T-test models:** Instead of regression models, difference of means t-tests were conducted to quantify the impact of Zearn Math on student achievement. Given SD =standard deviations and n =number of observations per group, t-tests were conducted as:

$$t = \frac{\text{mean}_{treatment} - \text{mean}_{control}}{\sqrt{\frac{SD^2_{treatment}}{n_{treatment}} + \frac{SD^2_{control}}{n_{control}}}} \quad (\text{Model 8})$$

In 2025, regression models were introduced to better control for residual confounding after robust matching, improving the precision of estimates and minimizing bias. This shift also enhanced the accuracy of measuring multi-grade-level samples for non-vertically scaled assessments by enabling the inclusion of grade-level dummies in the analytical models.

3. **Cohen's d:** Effect sizes were calculated with *Cohen's d*, which divides the difference in means between treatment and control by the pooled standard deviations:

$$Cohen's\ d = \frac{mean_{treatment} - mean_{control}}{pooled\ SD} \quad (Model\ 9)$$

The effect sizes in the revised methods use Hedge's *g*, a statistic similar to Cohen's *d*, but with a correction for small sample sizes that reduces bias in such cases.

Limitations

Zearn's approach to impact research utilizes quasi-experimental methods to control for observed confounders and zoom in on the relationship between Zearn Math use and student achievement among consistent Zearn Math users. While these methods build promising evidence of Zearn Math's effectiveness, limitations to quasi-experimental analyses exist and cannot fully replicate experimental conditions.

Even with robust quasi-experimental methods, the accuracy of estimates is limited by the ability to account for all variables influencing selection into treatment and control groups. Unobserved confounders may mediate the relationship between Zearn Math use and student achievement, leading to biased results. While Coarsened Exact Matching (CEM) helps control for observed confounders, there is still the potential for unmeasured factors that distinguish students who use Zearn Math consistently from those with no usage. Because of this risk of unobserved bias, the highest rating the WWC assigns to QED studies is "Meets WWC Standards with Reservations" (2022).

District- and state-level analyses employ a TOT framework to assess the impact of consistent Zearn Math usage, rather than using an ITT framework, which would define the treatment group as all students with access to Zearn Math (McCoy, 2017). Zearn's open-access policy—where all students are rostered regardless of their intention to use the platform—precludes the use of an ITT approach. Since rostering data does not differentiate between students who actively engage with Zearn Math and those who do not, an ITT analysis would fail to capture the true impact of the intervention. In contrast, the TOT framework more precisely measures the intervention's effect by focusing on the students who actually use the platform.

Using consistent lesson completion of 1-1.9, 2-2.9, and 3+ lessons-per-week as the benchmarks for treatment means that estimates may be biased and exceed “typical usage.” In addition, focusing on on- or above-grade lesson completion as a usage metric means that students who struggle more and therefore complete fewer lessons, even with the same amount of time on the platform, will be systematically excluded from the analysis.

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